

17.1 Simple Harmonic Motion

Question Paper

Course	CIE A Level Physics
Section	17. Oscillations
Topic	17.1 Simple Harmonic Motion
Difficulty	Easy

Time allowed: 50

Score: /38

Percentage: /100

Question 1a

Define an oscillation.

[2 marks]**Question 1b**

Identify the correct definition by drawing lines between the properties of oscillations and their definition.

Properties of Oscillations	Definition of Properties
Displacement	The rate of change of angular displacement with respect to time
Amplitude	The distance of an oscillator from its equilibrium position
Angular Frequency	The time taken for one complete oscillation
Frequency	The maximum displacement of an oscillator from its equilibrium position
Time Period	The number of oscillations per unit time

[5 marks]

Question 1c

Define simple harmonic oscillation.

[3 marks]

Question 1d

Use the words in the box below to correctly label the diagram of an oscillating pendulum in Fig 1.1.

displacement of mass mass acceleration restoring force equilibrium position

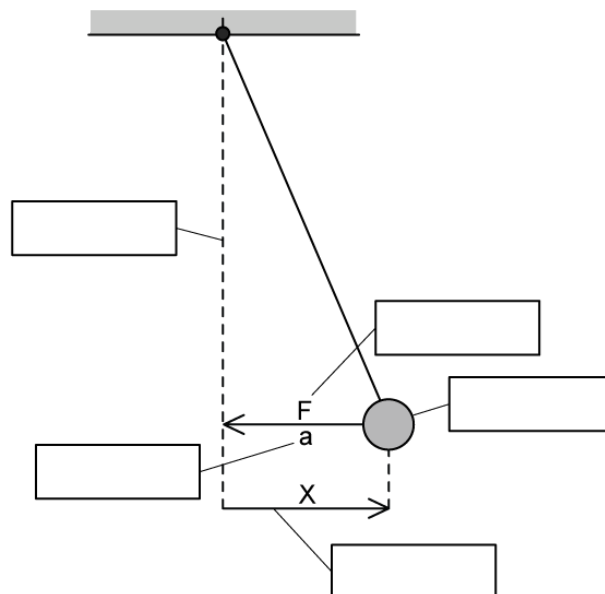


Fig. 1.1

[4 marks]

Question 2a

Identify by drawing a circle around the equation that defines simple harmonic motion and its two solutions.

$v = f\lambda$	$\omega = \frac{2\pi}{T}$	$x = x_0 \sin(\omega t)$
$x = x_0 \cos(\omega t)$	$a = -\omega^2 x$	$f = \frac{1}{T}$

[3 marks]

Question 2b

The graph in Fig. 1.1 shows that the acceleration of an object is directly proportional to the negative displacement.

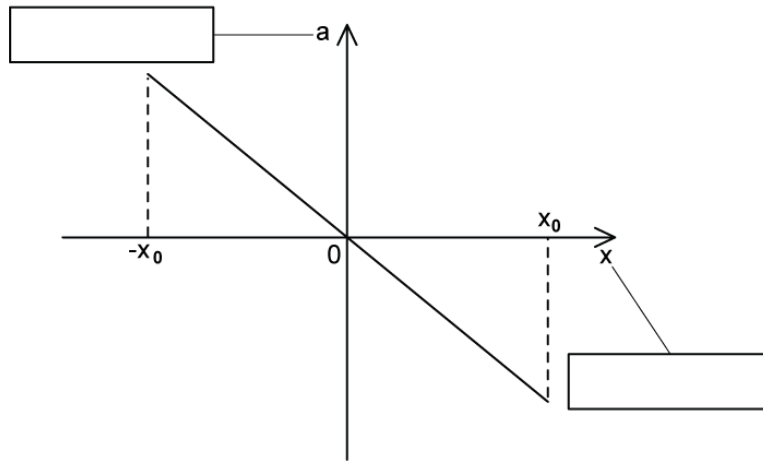


Fig. 1.1

Label the two axis of the graph on Fig. 1.1.

[2 marks]

Question 2c

The graphs in Fig. 1.3 show an oscillator starting at the equilibrium position and an oscillator starting at the maximum displacement.

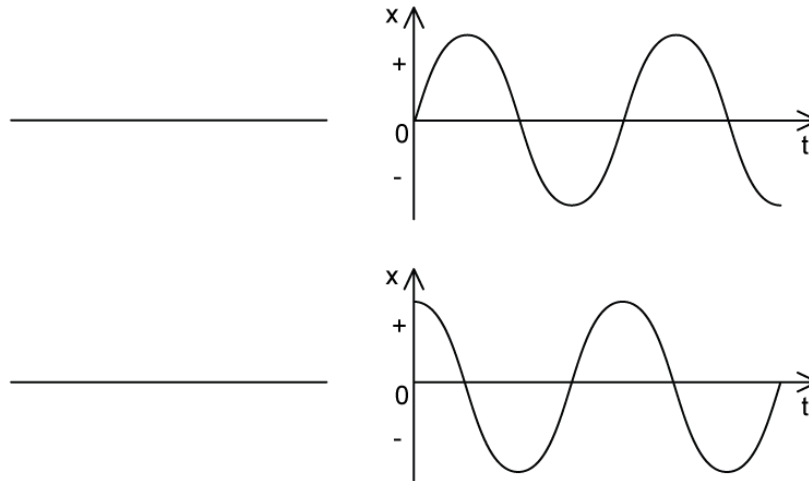


Fig. 1.3

Identify by writing next to the graphs on Fig 1.3 the correct name of the starting position of the oscillator.

[2 marks]

Question 2d

The equation below defines how the speed of an oscillator changes with the displacement of the oscillator.

$$v = \pm \omega \sqrt{(x_0 - x)}$$

Identify the variables in the equation by stating the variable and the quantity it represents in the space below.

[4 marks]

Question 3a

Complete the gaps, using the words from the box, in the sentences that describe the relationship between the simple harmonic motion graphs.

cosine sine identical motion displacement 45° 90° velocity
--

- The displacement-time graph is a curve identical to a _____ graph when the oscillator starts from its equilibrium position.
- Velocity is the rate of change of _____
- The velocity-time graph is _____ out of phase with the displacement-time graph
- Acceleration is the rate of change of _____
- The acceleration-time graph is _____ to the displacement graph except reflected on the x-axis.

[5 marks]

Question 3b

State the equation for the total energy of a simple harmonic system.

[1 mark]

Question 3c

Fig. 1.1 shows the graph of potential, kinetic and total energy of a simple harmonic system for half a period of simple harmonic oscillation.

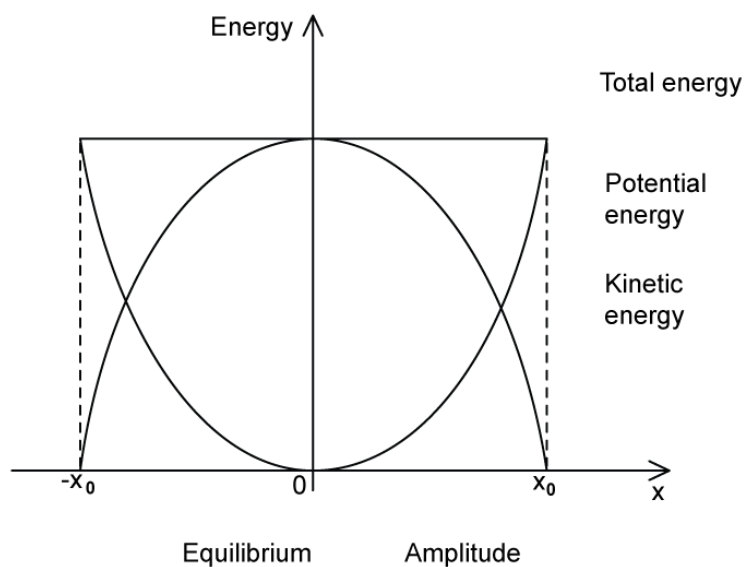


Fig. 1.1

Match the labels to the correct place on the graph by drawing a line between them.

[5 marks]

Question 3d

Identify, by placing a tick (✓) next to, the correct statements about the key features of the displacement-time graph.

Possible statements about the displacement-time graphs	Place a tick (✓) here if the statement is correct
The amplitude of oscillations x_0 can be found from the maximum value of x	
If the oscillations start at the positive or negative amplitude, the displacement will be at its minimum	
The time period of oscillations T can be found from reading the time taken for one full cycle	
The graph always starts at 0	

[2 marks]

Model Answers: Easy

1a

(a) An oscillation is:

- Repeated back and forth movements; [1 mark]
- On either side of any equilibrium position; [1 mark]

[Total: 2 marks]

1b

(b) Correctly drawn lines on the table that would score 5 marks should look like this:

Properties of Oscillations	Definition of Properties
Displacement	The rate of change of angular displacement with respect to time
Amplitude	The distance of an oscillator from its equilibrium position
Angular Frequency	The time taken for one complete oscillation
Frequency	The maximum displacement of an oscillator from its equilibrium position
Time Period	The number of oscillations per unit time

- One mark for each correct line drawn

[Total: 5 marks]

1c

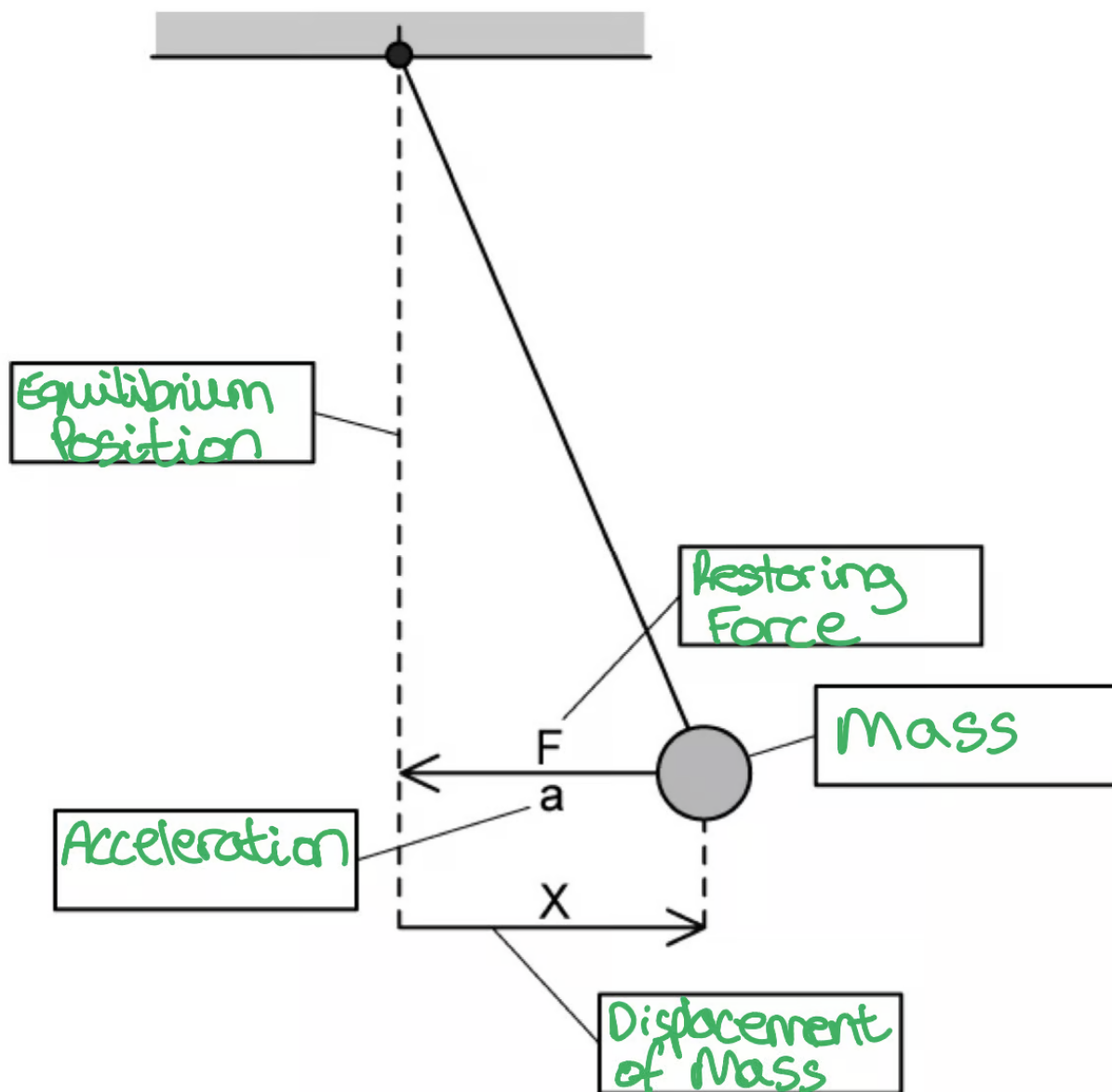
(c) Simple harmonic oscillation is:

- A type of oscillation in which the acceleration of a body; [1 mark]
- Is proportional to its displacement; [1 mark]
- But acts in the opposite direction; [1 mark]

[Total: 3 marks]

1d

(d) A correctly labelled diagram that would score 5 marks should look like this:



- One mark for each label in the correct place

[Total: 4 marks]

2a

(a) The equation that defines simple harmonic motion and its two solutions are:

A table that would score 3 marks should look like this:

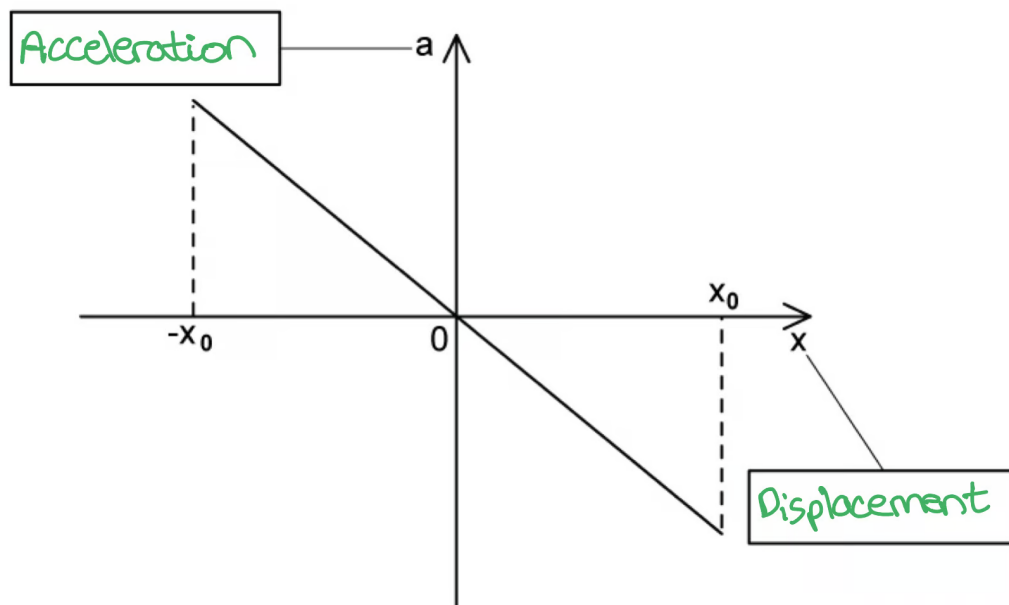
$v = f\lambda$	$\omega = \frac{2\pi}{T}$	$x = x_0 \sin(\omega t)$
$x = x_0 \cos(\omega t)$	$a = -\omega^2 x$	$f = \frac{1}{T}$

- One mark for each correct equation circled

[Total: 3 marks]

2b

(b) A graph that would score 2 marks should look like this:



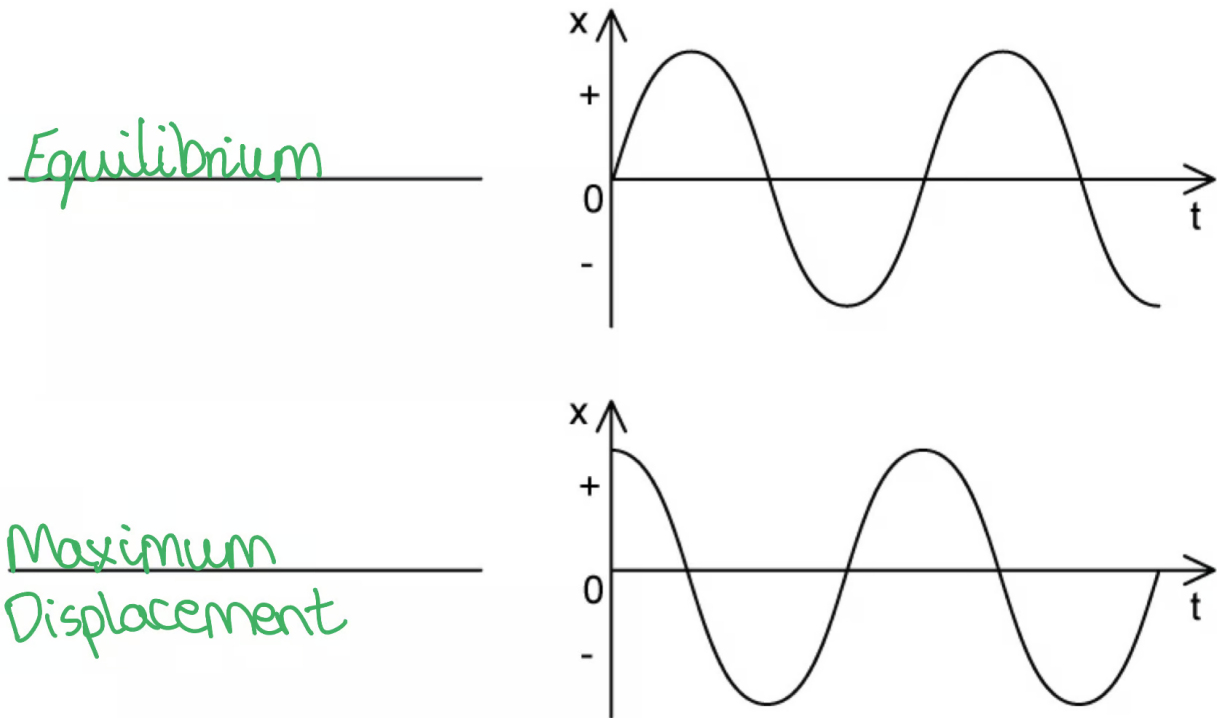
- Horizontal axis is labelled as displacement; **[1 mark]**

- Vertical axis is labelled as acceleration; [1 mark]

[Total: 2 marks]

2c

(c) Two correctly labelled graphs that would score 2 marks should look like this:



- Top graph labelled as starting in the equilibrium position; [1 mark]
- Bottom graph labelled as starting at the maximum displacement; [1 mark]

[Total: 2 marks]

2d

(d) Identify the variables in the equation:

- v = speed of oscillator; [1 mark]
- ω = angular frequency of oscillation; [1 mark]
- x_0 = amplitude of oscillation; [1 mark]
- x = displacement of oscillator; [1 mark]

[Total: 4 marks]

3a

(a) Sentences completed with the correct words:

- The displacement-time graph is a curve identical to a **sine** graph when the oscillator starts from its equilibrium position; [1 mark]
- Velocity is the rate of change of **displacement**; [1 mark]
- The velocity-time graph is **90°** out of phase with the displacement-time graph; [1 mark]
- Acceleration is the rate of change of **velocity**; [1 mark]
- The acceleration-time graph is **identical** to the displacement graph except reflected on the x-axis; [1 mark]

[Total: 5 marks]

3b

(b) The equation for the total energy of a simple harmonic system is:

- $E = \frac{1}{2}m\omega^2(x_o)^2$ [1 mark]

[Total: 1 mark]

3c

(c) A diagram that would score 5 marks should look like this:

- The velocity-time graph is **90°** out of phase with the displacement-time graph; [1 mark]
- Acceleration is the rate of change of **velocity**; [1 mark]
- The acceleration-time graph is **identical** to the displacement graph except reflected on the x-axis; [1 mark]

[Total: 5 marks]

3b

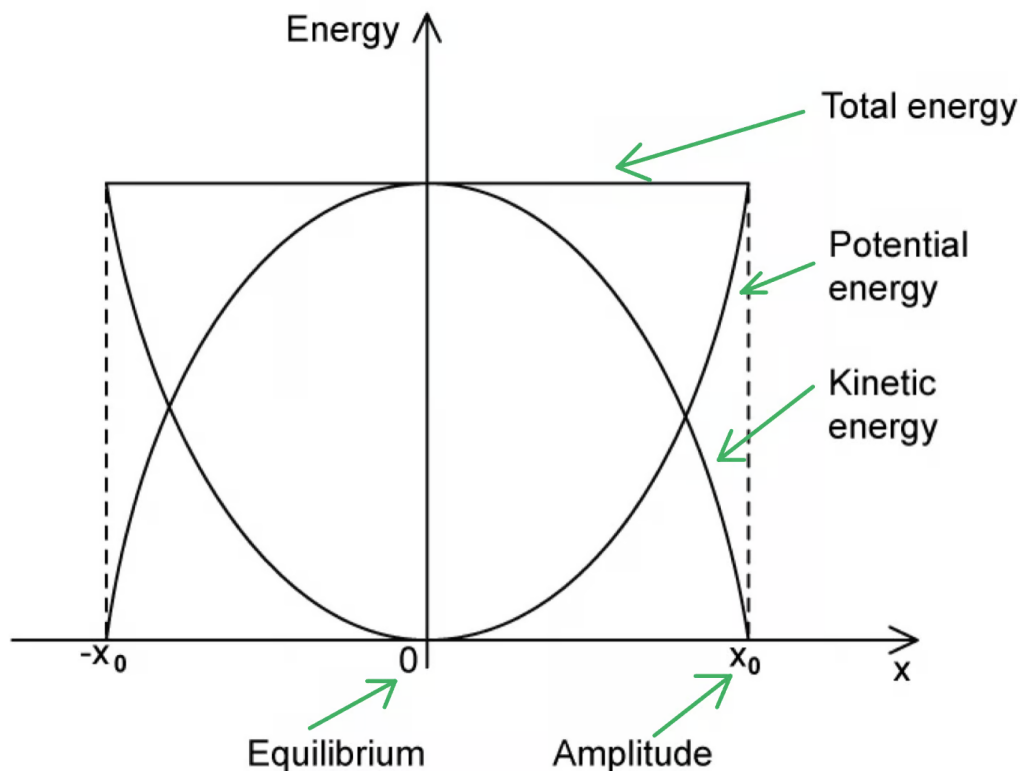
(b) The equation for the total energy of a simple harmonic system is:

- $E = \frac{1}{2}m\omega^2(x_o)^2$ [1 mark]

[Total: 1 mark]

3c

(c) A diagram that would score 5 marks should look like this:



- Total energy joined to horizontal axis; [1 mark]
- Potential energy joined to U shaped curve; [1 mark]
- Kinetic energy joined to n shaped curve; [1 mark]
- Equilibrium joined to 0 on horizontal axis; [1 mark]
- Amplitude joined to one or both of x_0 or $-x_0$ on the horizontal axis; [1 mark]

[Total: 5 marks]

3d

(d) A table that would score 2 marks should look like this:

Possible statements about the displacement-time graphs	Place a tick (✓) here if the statement is correct
The amplitude of oscillations x_0 can be found from the maximum value of x	✓ [1 mark]
If the oscillations start at the positive or negative amplitude, the displacement will be at its minimum	
The time period of oscillations T can be found from reading the time taken for one full cycle	✓ [1 mark]
The graph always starts at 0	

- One mark for each correct statement ticked

[Total: 2 marks]

The two incorrect statements are correct below:

- If the oscillations start at the positive or negative amplitude, the displacement will be at its **maximum**
- The graph **might not** always starts at 0

